Earth's Orbital Speed from Three Radii: $v_\oplus \approx \dfrac{R_\odot \times R_{\rm Moon}}{R_\oplus^2}$

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Abstract

We report an empirical numerical coincidence. For the Sun, Earth, and Moon:

$$v_\oplus \approx \frac{R_\odot \times R_{\rm Moon}}{R_\oplus^2}$$

With IAU mean radii, the right-hand side evaluates to 29.78. The same number (to within 0.013%) equals Earth's mean orbital speed in km/s. The relation uses only lengths — no masses, no gravitational constant, no Kepler laws. We present it as an empirical geometric observation within Earth-based units.

1 Numerical Evaluation

Using IAU 2026 mean radii:

$$R_\odot = 695\,700.0 \text{ km}$$
$$R_{\rm Moon} = 1\,737.4 \text{ km}$$
$$R_\oplus = 6\,371.0 \text{ km}$$

$$R_\odot R_{\rm Moon} = 1.20870918 \times 10^9 \text{ km}^2, \qquad R_\oplus^2 = 4.05896410 \times 10^7 \text{ km}^2.$$

$$\frac{R_\odot R_{\rm Moon}}{R_\oplus^2} \approx 29.7788$$

The Earth's mean orbital speed (IAU) is $v_\oplus^{\rm IAU} = 29.7827$ km/s. Relative gap: 0.013%.

2 Why Units Are Compatible

The kilometer is a fraction of Earth's circumference (1/40 075). The second is a fraction of Earth's rotation period (1/86 400). In a consistent Earth-based unit system (time expressed as length via equatorial rotation), velocity becomes dimensionless, and the relation reduces to a geometric statement about three radii.

3 Conclusion

Within Earth-based units, the three radii of the Sun, Earth, and Moon satisfy

$$v_\oplus \approx \frac{R_\odot \, R_{\mathrm{Moon}}}{R_\oplus^2}$$

to 0.013% precision. No masses, gravitational constant, or Kepler laws are used. Whether this coincidence has a deeper geometric origin or remains a numerical curiosity is left open.

See also (Zenodo)

1. **Platonic G: Gravitation from Geometry**
   https://zenodo.org/records/20009102

2. **Kochenov Geometric Identity for G**
   https://zenodo.org/records/19769041

3. **Geometric Origin of 10^{11} in the Gravitational Constant (Corrected)**
   https://zenodo.org/records/20009474

4. **G from Earth's Geometry** (May 2026)

Keywords: orbital speed, Earth, Sun, Moon, geometric coincidence, Earth-based units, parts of the Earth